

TASK 6

Implement DES-2 and DES-3 using Java cryptography package

Aim: To implement DES-2 and DES-3 using Java cryptography package

Program:

```
import java.security.*;
import java.io.*;
import javax.crypto.*;
import java.util.*;
public class Task6 {
    static Scanner s = new Scanner(System.in);
    private static Cipher cipher;
    private static byte[] encrypt(String data, SecretKey key1, SecretKey key2, int flag) throws
Exception {
        String algo = (flag == 2) ? "DESede" : "DES";
        cipher = Cipher.getInstance(algo +
"/ECB/PKCS5Padding");
        cipher.init(Cipher.ENCRYPT_MODE, key1);
        byte[] encrypted = cipher.doFinal(data.getBytes());

        if (flag == 1) {
            cipher.init(Cipher.ENCRYPT_MODE,
            key2); encrypted =
            cipher.doFinal(encrypted);
        }
        System.out.println("Encrypted: " + new
String(encrypted)); return encrypted;
    }
    private static String decrypt(byte[] data, SecretKey key1, SecretKey key2, int flag) throws
Exception {
        String algo = (flag == 2) ? "DESede" :
"DES"; byte[] decrypted;
        cipher = Cipher.getInstance(algo + "/ECB/PKCS5Padding");
        if (flag == 1) {
            cipher.init(Cipher.DECRYPT_MODE,
            key2); decrypted =
            cipher.doFinal(data);
        } else {
            decrypted = data;
        }
        cipher.init(Cipher.DECRYPT_MODE,
        key1); decrypted =
        cipher.doFinal(decrypted);
        System.out.println("Decrypted: " + new String(decrypted));
        return new String(decrypted);
    }
    public static void main(String[]
args) { try {
```

```

System.out.print("Enter 1 for 2DES or 2 for
3DES: "); int flag = s.nextInt();
KeyGenerator keygen = KeyGenerator.getInstance((flag == 2) ? "DESede" :
"DES"); SecretKey key1 = keygen.generateKey();
SecretKey key2 = keygen.generateKey();
byte[] encrypted = encrypt(data, key1, key2, flag);
decrypt(encrypted, key1, key2, flag);
} catch (Exception e)
{
    e.printStackTrace();
}
}
}

```

Apache NetBeans IDE 14

Inspection 2025

Home Desktop Inspection 2025

Places

Recent

Home

Desktop

Downloads

Music

Pictures

Videos

Trash

Devices

31 GB Volume

Computer

Network

Browse Network

Connect to S...

CaesarCipher11 - Apache NetBeans IDE 14

File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help

Search (Ctrl+I)

Project: ... Services Files

Task11.java

run.xml

Output

Task11 (run) #3

CaesarCipher11 (run) #4

CaesarCipher11 (run) #5

CaesarCipher11 (run) #6

run:

Enter plaintext: gokaraju

Enter 1 for 2DES or 2 for 3DES: 1

Encrypted: 9p0000200000hy

Decrypted: gokaraju

BUILD SUCCESSFUL (total time: 22 seconds)

- 1.What is the main difference between DES, Double DES (DES-2), and Triple DES (DES-3)?
- 2.Why is Double DES (DES-2) vulnerable to the meet-in-the-middle attack?
- 3.How does Triple DES (DES-3) improve security over DES-2?
4. What are the different modes of operation for Triple DES (DES-3)?